WILLIAM HUANG

william.huang@ucla.edu | whuang37.github.io | whuang37 | whuang37

EDUCATION

University of California, Los Angeles September 2024 – Present
Ph.D. in Electrical and Computer Engineering (GPA: 4.00/4.00) *Los Angeles, CA*
Thesis: *Towards General Embodied User Models*

University of California, Los Angeles September 2024 – June 2025
M.S. in Electrical and Computer Engineering (GPA: 4.00/4.00) *Los Angeles, CA*
Thesis: *Data Synthesis Techniques to Improve Pose Estimation Performance on Wheelchair Users*

University of California, Los Angeles September 2020 – June 2024
B.S. in Computer Engineering, *Cum Laude* (GPA: 3.92/4.00) *Los Angeles, CA*

PROFESSIONAL EXPERIENCE

University of California, Los Angeles December 2021 – Present
Graduate Student Researcher, *Advised by Prof. Yang Zhang* *Los Angeles, CA*
– Design human-AI systems for embodied preference elicitation and simulations alignment/reasoning.

Google June 2026 – December 2026
Student Researcher, *Advised by Dr. Ruofei Du, Dr. David Li* *San Francisco, CA*
– Benchmarking AI's understanding of extended reality development through embodied user simulations.

Pacific Northwest National Laboratory June 2023 – December 2023
Applied Decision Systems and Analytics Intern *Virtual / Richland, WA*
– Optimized big data processing of always-on ammonia farm sensor arrays through high performance computing.

University of California, Los Angeles February 2023 – December 2023
Research Assistant, *Advised by Prof. Vwani P. Roychowdhury* *Los Angeles, CA*
– Reduced training time of motion generation models through GPU optimization and biomechanical modeling.

Intel Corporation January 2022 – September 2022
Biomechanics Co-op *Virtual / Santa Clara, CA*
– Designed, trained, and implemented pose estimation applications in athletics training and rehabilitation.

UCLA Athletics December 2020 – January 2022
Data Analytics and Sports Science Project Lead *Los Angeles, CA*
– Led a team of 20 to develop end-to-end data pipelines and visualizations to manage athlete wellness data.

National Institutes of Health – Office of AIDS Research June 2021 – August 2021
Data Science Fellow *Virtual / Bethesda, MD*
– Developed an automated grant proposal classification pipeline to analyze and assign reviewers.

University of California, Los Angeles September 2020 – July 2021
Research Assistant, *Advised by Prof. Xiang 'Anthony' Chen* *Los Angeles, CA*
– Studied predictors of problem gambling behavior in daily fantasy sports activity.

University of California, Irvine June 2020 – June 2021
Research Assistant, *Advised by Prof. Edwin S. Monuki* *Irvine, CA*
– Developed human-centered pathology annotation tools to reduce the physical strain of data collection.

PUBLICATIONS

William Huang, Siyou Pei, Leyi Zou, Eric J Gonzalez, Ishan Chatterjee, and Yang Zhang. "DeltaDorsal: Enhancing Hand Pose Estimation with Dorsal Features in Egocentric Views". In: *Proceedings of the 2026 CHI Conference on Human Factors in Computing Systems*. CHI '26. Barcelona, Spain: Association for Computing Machinery, 2026. ISBN: 9798400722783. DOI: 10.1145/3772318.3790493.

William Huang, Xia Su, Jon E. Froehlich, and Yang Zhang. "Accessibility Scout: Personalized Accessibility Scans of Built Environments". In: *The 38th Annual ACM Symposium on User Interface Software and Technology (UIST '25), September 28-October 1, 2025, Busan, Republic of Korea*. UIST '25. New York, NY, USA: Association for Computing Machinery, Sept. 2025. ISBN: 9798400720376. DOI: 10.1145/3746059.3747624.

William Huang, Sam Ghahremani, Siyou Pei, and Yang Zhang. "WheelPose: Data Synthesis Techniques to Improve Pose Estimation Performance on Wheelchair Users". In: *Proceedings of the CHI Conference on Human Factors in Computing Systems*. CHI '24. New York, NY, USA: Association for Computing Machinery, May 2024, pp. 1–25. ISBN: 979-8-4007-0330-0. DOI: 10.1145/3613904.3642555.

Deborah K. Fagan, Anastasia Bernat, Brett C. Simpson, Randy K. Tran, John Wilson, **William Huang**, Clara L. Stickney, David LePoire, Mark MacDonald, R. D. Hildebrand, and Hoffman Edward. "Sample Planning for Transportation Accident Environmental Characterization Using RISKIND and VSP". In: *Proceedings of the Waste Management Symposia*. WMS 2024. Phoenix, AZ, USA: Waste Management Symposia, Mar. 2024.

POSTERS, PREPRINTS, EXTENDED ABSTRACTS

William Huang, David Li, Xiang 'Anthony' Chen, Xun Qian, Nels Numan, Zhongyi Zhou, Hyunsung Cho, Salman Muin Kayser Chishti, Ding Xia, Xincheng Huang, Jiahao Ren, Faraz Faruqi, Tianyu Xu, Yang Zhang, David Kim, and Ruofei Du. *GenXR-Bench: Benchmarking Coding Agents for Interactive XR Development*. Oct. 2026. DOI: 10.48550/arXiv.2610.29014.

William Huang, Ruofei Du, and Yang Zhang. *AlterAtlas: Shifting Travel Planning from AI Generation to Validation via Persona-Driven Simulations*. July 2026. DOI: 10.48550/arXiv.2607.16565.

N. Victoria Espericueta, Michael J. Neel, Yao Wang, Todd J. Soo, Parastou Porahang, Frances A. Goyokpin, Ryan S. Salehi, Sara Khan, Jasmine Lee, Nicolas B. Maramica, Guillermo Flores, Aditi Kulkarni, Sahil S. Plaha, Sam Ghahremani, **William Huang**, Quinton Smith, Peter Chang, Brett A. Johnson, and Edwin S. Monuki. *Inducible Lipid Storage and Steatosis in the Human Choroid Plexus Associated with Age and Adiposity*. May 2026. DOI: 10.64898/2026.05.21.725559.

Jon E. Froehlich, Jared Hwang, Zeyu Wang, John S. O'Meara, Xia Su, **William Huang**, Yang Zhang, Alex Fiannaca, Philip Nelson, and Shaun Kane. "'Does the Cafe Entrance Look Accessible? Where Is the Door?' Towards Geospatial AI Agents for Visual Inquiries". In: *ICCV Workshop on Vision Foundation Models and Generative AI for Accessibility: Challenges and Opportunities (CV4A11y)*. arXiv, Aug. 2025. DOI: 10.48550/arXiv.2508.15752.

SELECTED HONORS AND AWARDS

NSF GRFP ($37,000 per year, 3 years) 2025
QWER Hacks Third Place, Education Track 2021
Beckman Legacy Awardee ($8,000) 2020

PAPER REVIEWING

ACM CHI 2026, 2027
ACM UIST 2025, 2026
ACM TACCESS 2026
ACM IMWUT November 2025, February 2026
ACM CHI LBW 2025

SELECTED OUTREACH

Idea Reviewer, Project Invent 2026
Assistant to SC, ACM CHI 2026 2025

Mentor, Coding it Forward 2025
HCC and AI/ML Instructor and Organizer, Los Angeles Computing Circle 2025
English Instructor, Independent 2020 – 2025
Sports Analytics Outreach Head, UCLA DataRes 2021 – 2022
Application Reviewer, Coding it Forward 2021 – 2022
Curriculum Developer and Python Instructor, ACM Teach LA 2020 – 2021

SELECTED TALKS AND PRESENTATIONS

Florida International University, *Towards General Embodied User Models* 2026
Carnegie Mellon University, *Towards General Embodied User Models* 2026

MENTORSHIP

Humad Bari January 2026 – Present
Linguistics and Computer Science at UCLA

Leyi Zou September 2024 – May 2026
Next – Computer Science PhD at Purdue University

Alexandra Villanueva June 2025 – January 2026
Electrical Engineering Undergraduate at UCLA

Medha Kini September 2024 – June 2025
Next – Software Engineer at Google

Tracy Zhao June 2024 – June 2025
Next – Software Engineer at Google

Brandon Yan June 2024 – June 2025
Next – Software Engineer at Atlassian

Sam Ghahremani March 2023 – August 2024
Next – Human-Centered Computing PhD at the Georgia Institute of Technology